

Backtracking: Bloodline

Game Design

Version 1.0.0

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1.Game overview:

Title: *Time Escape VR*

Genre: VR Puzzle / Escape Room

Platform: PC VR (Unity-based)

Target Audience: Puzzle lovers, VR beginners, Unity learners

What it's about: **Backtracking:Bloodline VR** is a first-person virtual reality escape room game where the player wakes up in a mysterious secret laboratory. The player must solve four interactive puzzles to escape—two in the starting room and two in the second room, which is only accessible after powering up the lab and acquiring a key card. The twist lies in the visual and thematic transition between the two rooms, each designed to reflect a different historical era, emphasizing the concept of time travel. The experience is designed to be short, engaging, and accessible to new VR users.

2.Unique Selling Points:

- **Immersive First-Person VR Experience:** Players interact with the world directly using VR controls such as grabbing, pressing and rotating.
- **Time Travel Theme:** Each room is styled after a different era. Room 1 is a high-tech modern lab, while Room 2 has a World War-style office aesthetic.
- **Simple Yet Effective Storyline:** The absence of heavy narration or complex plot allows players to focus entirely on exploration and puzzle solving.
- **Beginner-friendly:** Straightforward mechanics and story make it ideal for VR newcomers and Unity learners.
- **Educational and Accessible:** Ideal for students or beginner Unity users interested in learning VR development and puzzle design.

3.Game Loop:

1. **Wake up in Room 1:** The player starts disoriented and need to look around and explore their surroundings.
2. **Solve Puzzle 1:** An environmental logic puzzle, the player needs to repair the circuit with a simple game. The circuit will be highlighted as the room will be a bit dark before restarting power.
3. **Solve Puzzle 2:** Follow the tips to find the access card to Room 2 and probably a pistol. For this part the player might need to move and rotate the objects as things might be put in the pockets of a lab coat.
4. **Access Room 2:** Using the key card, the player unlocks the door to the next room.

5. **Enter Room 2:** Visually and thematically distinct from Room 1 (war-era office). The two rooms are separate and user will be teleported to room 2 after the door of room 1 is opened.
6. **Solve Puzzle 3:** Interact with equipments in the room to open a safe box using brute force
7. **Solve Puzzle 4:** Final challenge before escape, open the hidden door with the documents behind it. This part players will interact with a keypad and figure out the password.
8. **Return to Room 1 and Escape:** The player can now walk out of room 1 and the game ends.

4.Gameplay Scope:

- **Total Rooms:** 2
- **Total Puzzles:** 4 (2 per room)
- **Completion Time:** 10-20 minutes
- **Replayability:** Low to medium; can be replayed for speed or exploring different puzzle strategies
- **Game Mode:** Single-player only
- **Difficulty Level:** Easy to medium – designed for accessibility
- **Controls:** Grab/release items, Push buttons/switches, Rotate or slide puzzle components, Optional: shoot using VR trigger

5.Art Style:

- **Visual Direction:** Clean, stylized low-poly assets with distinct visual clues for interactable objects.
- **Room 1 (Modern Lab):**
 - Sleek, high-tech environment with screens and glass materials.
 - Neon blue/white lights; minimalistic sci-fi design.
- **Room 2 (War Period Office):**
 - Relatively vintage decor, primarily wooden furniture.

- Interactive items: office desk, drawer, safe that are highlighted in bright colors(yellow and red).
- **User Interface:**
 - Floating 3-folded menu with all the basic functions(including backpack with the keys stored) that allows players to operate without clicking for a new page.
 - Diegetic UI(puzzles or clues embedded in the environment).
- **Performance:** Optimized for Unity using basic lighting, simplified geometry and efficient materials.

6.Player Profiles

1. **VR beginners:** This game provides a beginner-friendly experience with intuitive controls and short playtime. It's ideal for casual gamers or those experimenting with VR for the first time.
2. **Unity VR Developers / Students:** Unity learners can use the project as a base to practice puzzle scripting, interaction design, and scene management.
3. **Puzzle Game Enthusiasts:** Players who enjoy escape rooms or classic puzzle-solving will appreciate the balance between exploration and logic.
4. **Player Motivation:**

Curiosity: Discover how the rooms and puzzles are connected.

Problem-solving: Enjoy figuring out clues using visual hints.

Escapism: A short VR experience to travel through time in under 20 minutes.

7. Milestone Schedule

Date	Milestone	Status
6 July	Concept finalized, initial GDD drafted	Done
10 July	Room 1 setup + Puzzle 1 created	In Progress
17 July	Room 2 scene built + Puzzles designed	Pending
25 July	Interactions tested (grab, shoot, press)	Upcoming

Date	Milestone	Status
1 August	Final build + polishing + submission	Upcoming

8. Project Vitals

- **Engine:** Unity 2022 LTS
- **Toolkit:** XR Interaction Toolkit
- **VR Hardware:** Any VR headsets supporting android system
- **Assets:** Unity Asset Store
- **Development Time:** 1 month
- **Team Size:** Solo(individual project)
- **Version Control:** Git(local backup and file management)
- **Challenges:** Unity VR physics bugs, optimizing performance for smooth frame rate, implementing shooting mechanic

9. Backstory

The player is a scientist with special abilities that can backtrack to his ancestors' era to observe historical events from their sights and can bring back one thing to his own era. Recently, the lab started a project about human experiments that has been initiated during war period. Since human experiments are inhumane, the player needs to backtrack to one of his ancestors who's an agent during the war and stole the experiment documents successfully to bring it back to promote the progress of the project and seek opportunities to perform experiments on animals for same outcomes.

10.Flowchart

